

APPLICATION OF DERIVATIVES

• **Rate Measure:**

(1) Let $y = f(x)$ be a function defined on neighborhood of x . If Δx is the small change in x , and the corresponding change in y be Δy then $\frac{\Delta y}{\Delta x}$ is called average change of $y = f(x)$ in $[x, x + \Delta x]$.

(a) If $\frac{\Delta y}{\Delta x}$ is constant then we say that $y = f(x)$ changes at a uniform rate wrt x .

(b) If $\frac{\Delta y}{\Delta x}$ is not constant then we say that $y = f(x)$ changes at a variable rate wrt

x . If $\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$ exist then the limit is called as rate of change of y wrt x and is denoted by $\frac{dy}{dx}$.

(2) **Velocity:** If a particle moves along a line by the relation $s = f(t)$, then velocity $v = f'(t) = \frac{ds}{dt}$ of the particle at t .

(a) If $v > 0$ then s is increasing and the particle is moving away from the point of observation.

(b) If $v < 0$ then s is decreasing and the particle is moving towards the point of observation.

(c) If $v = 0$ then the particle comes to rest.

(3) **Acceleration:** The rate of change in velocity is called as acceleration of the particle at t and is denoted by a . Here $a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$ and $a = \frac{dv}{dt} = \frac{dv}{ds} \frac{ds}{dt} = v \frac{dv}{ds}$.

(4) **Angular Velocity & Acceleration:** Let O be the fixed point and OX be the fixed ray. Let P be the position of the particle on a curve C at time t such that $\angle XOP = \theta$.

(a) Angular velocity $= \omega = \frac{d\theta}{dt}$.

(b) Angular acceleration $= \alpha = \frac{d^2\theta}{dt^2}$.

(5) **Rate of Change:**

(a) If both x and y are functions of the parameter t , then $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$.

(b) If the rate of change of a variable is positive (negative) then the value of the variable increases (decreases) with the increase in the value of the independent variable.

(6) **Errors & Approximation:**

- (a) If $y = f(x)$ and Δx is the small change in x then the corresponding approximate change in y is given by $\Delta y = f'(x)\Delta x$. This is called the differential of y and is denoted by $dy \approx f'(x)\Delta x$.
- (b) The actual error in y is denoted by $\Delta y = f(x + \Delta x) - f(x)$.
- (c) $dy \approx \Delta y \approx f'(x)\Delta x$.
- (d) If Δx is the error in x then the approximate value of $f(x)$ is $f(x + \Delta x) \approx f(x) + f'(x)\Delta x$.
- (e) Let Δx be any change in x then and Δy be the corresponding change in y . Then:
 1. Δy is called absolute error in y .
 2. $\frac{\Delta y}{y}$ is called as relative error in y .
 3. $\frac{\Delta y}{y} \times 100$ is called as percentage error in y .
- (f) If $f(x) = ax^2 + bx + c$, then $\Delta f - df = a(\Delta x)^2$.
- (g) If $y = f(x)$ is a homogeneous function of degree n , then
 1. Relative error in $y = n$ [Relative error in x].
 2. Percentage error in $y = n$ [Percentage error in x].

• **Tangent & Normal:**

- (1) The slope of tangent to the curve $y = f(x)$ at point (x_1, y_1) is $\left. \frac{dy}{dx} \right|_{(x_1, y_1)}$.
- (2) The slope of the tangent to curve at a point is called as gradient to the curve at that
- (3) A line which is perpendicular to the tangent of the curve at point P and passes through P is called a normal to the curve at P .
- (4) The slope of normal to the curve $y = f(x)$ at point (x_1, y_1) is $\left. \frac{-1}{\frac{dy}{dx}} \right|_{(x_1, y_1)}$.
- (5) The equation of tangent to the curve $y = f(x)$ at point (x_1, y_1) is:

$$(y - y_1) = \left. \frac{dy}{dx} \right|_{(x_1, y_1)} (x - x_1)$$
- (6) If $\left. \frac{dy}{dx} \right|_{(x_1, y_1)} = 0$, then the tangent to the curve $y = f(x)$ at point (x_1, y_1) is $(y - y_1) = 0 \Rightarrow y = y_1$ which is parallel to x -axis and the equation of normal to the curve $y = f(x)$ at point (x_1, y_1) is $(x - x_1) = 0 \Rightarrow x = x_1$ which is parallel to y -axis.

- (7) If $\left. \frac{dx}{dy} \right|_{(x_1, y_1)} = 0$, then the tangent to the curve $y = f(x)$ at point (x_1, y_1) is $(x - x_1) = 0 \Rightarrow x = x_1$ which is parallel to y-axis and the equation of normal to the curve $y = f(x)$ at point (x_1, y_1) is $(y - y_1) = 0 \Rightarrow y = y_1$ which is parallel to x-axis.
- (8) If $\left. \frac{dy}{dx} \right|_{(x_1, y_1)} \neq 0$ then the equation of normal to the curve $y = f(x)$ at point (x_1, y_1) is:
- $$(y - y_1) = \frac{-1}{\left. \frac{dy}{dx} \right|_{(x_1, y_1)}} (x - x_1)$$
- (9) The tangent at any point of the curve $x = at^3, y = at^4$ divides the abscissa of the point of contact in the ratio 1:3.
- (10) Point on the curve $ay^2 = x^3$ the normal at which makes equal intercepts on the axes is $\left(\frac{4a}{9}, \frac{8a}{27} \right)$.
- (11) The condition for the line $y = mx + c$ to be tangent to the parabola $y^2 = 4ax$ is $c = \frac{a}{m}$.
- (12) The equation of common tangent to the parabolas $y^2 = 4ax$ and $x^2 = 4by$ is $a^{\frac{1}{3}}x + b^{\frac{1}{3}}y + (ab)^{\frac{2}{3}} = 0$.
- (13) The condition for the line $y = mx + c$ to be tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $c^2 = a^2m^2 + b^2$.
- (14) The condition for the line $y = mx + c$ to be tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $c^2 = a^2m^2 - b^2$.

• **Angle of Intersection of Two Planes Curves:**

- (1) The angle between two curves is defined as the angle between the two tangents at their common point of intersection.
- (2) Let $y = f(x)$ and $y = g(x)$ be two plane curves intersect at (x_1, y_1) . If $m_1 = f'(x_1)$ and $m_2 = g'(x_1)$ are the slopes of tangent at (x_1, y_1) .
- (a) Acute angle between them $\theta = \tan^{-1} \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$.
- (b) Obtuse angle between them $\theta = \pi - \tan^{-1} \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$.

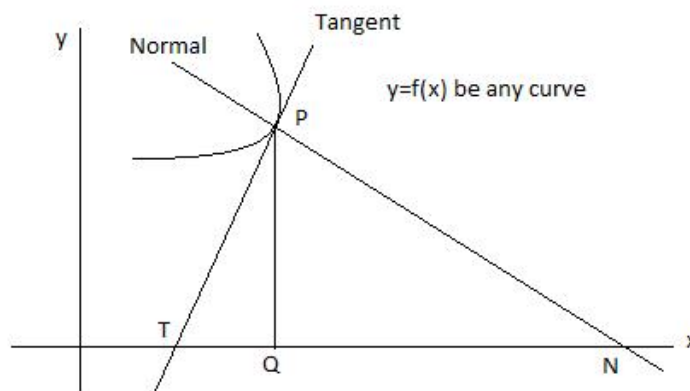
- (3) If $m_1 m_2 = -1$, then the two curves intersect orthogonally at the point of intersection.
 (4) If $m_1 = m_2$, then two curves touch each other at (x_1, y_1) and have common tangent.
 (5) If the curves $y^2 = 4ax$ and $x^2 = 4ay$ intersect at $(0, 0)$, $(4a, 4a)$. At $(0, 0)$ the curves cut each other orthogonally and at $(4a, 4a)$ the angle between the curves is $\tan^{-1} \frac{3}{4}$

then area between the two curves is $\frac{16a^2}{3}$.

- (6) If the curves $xy = c^2$ and $y^2 = 4ax$ cut each other orthogonally then $c^4 = 32a^4$.
 (7) The condition that two curves $x = y^2$ and $xy = k$ cut each other orthogonally is $8k^2 = 1$.
 (8) If the curves $a_1 x^2 + b_1 y^2 = 1$ and $a_2 x^2 + b_2 y^2 = 1$ cut each other orthogonally, then $\frac{1}{a_1} - \frac{1}{a_2} = \frac{1}{b_1} - \frac{1}{b_2}$.
 (9) Angle between the curves $y^2 = 4ax$ and $x^2 = 4by$ at their point of intersection is

$$= \tan^{-1} \left(\frac{3a^{\frac{1}{3}} b^{\frac{1}{3}}}{2 \left(a^{\frac{2}{3}} + b^{\frac{2}{3}} \right)} \right).$$

• **Length of Tangent, Normal, Sub-Tangent and Sub-Normal:**



- (1) Let the tangent at P to the curve meet x-axis in T and the normal meet x-axis in N. PQ perpendicular drawn to x-axis. Then
 (a) PT is called the length of tangent at P.
 (b) PN is called the length of normal at P.
 (c) QT, the projection of PT on x-axis called as length of sub-tangent.
 (d) QN, the projection of PN on x-axis called as length of sub-normal.

- (2) Let $P(x_1, y_1)$ be a point on the curve $y = f(x)$ and $m = \left. \frac{dy}{dx} \right|_{(x_1, y_1)}$. Then
- Length of tangent at P = $\left| \frac{y_1}{m} \right| \sqrt{1 + m^2}$.
 - Length of normal at P = $|y_1| \sqrt{1 + m^2}$.
 - Length of sub-tangent = $\left| \frac{y_1}{m} \right|$.
 - Length of sub-normal at P = $|y_1 m|$.
- (3) The area of triangle formed by the tangent at a point $P(x_1, y_1)$, normal at $P(x_1, y_1)$ and x-axis is $y_1^2 \frac{1 + m^2}{2|m|}$ sq units.
- (4) The area of triangle formed by the tangent at a point $P(x_1, y_1)$, normal at $P(x_1, y_1)$ and y-axis is $x_1^2 \frac{1 + m^2}{2|m|}$ sq units.
- (5) The area of triangle formed by the tangents at a point $P(x_1, y_1)$ on $y = f(x)$ with co-ordinate axes is $\frac{(y_1 - mx_1)^2}{2|m|}$ sq units.
- (6) The area of triangle formed by the normals at a point $P(x_1, y_1)$ on $y = f(x)$ with co-ordinate axes is $\frac{(x_1 + my_1)^2}{2|m|}$ sq units.
- (7) For the curve $x^n + y^n = a^n$, $P(x, y) = (a \cos^{2/n}, a \sin^{2/n})$ is a point on the curve.
- (8) The length of the sub-normal at any point on the curve $xy^n = a^{n+1}$ is a constant then $n = -2$.
- (9) For the curve $by^2 = (x+a)^3$, $\frac{(L.S.T)^2}{L.S.N} = \frac{8b}{27}$, where L.S.T – Length of sub-tangent and L.S.N – Length of sub-normal.

• **Rolle's Theorem:** If a function $f[a, b] \rightarrow R$ is such that:

- It is continuous on $[a, b]$.
- It is derivable on (a, b) .
- $f(a) = f(b)$

Then there exist at least one $c \in (a, b)$ such that $f'(c) = 0$.

OR

- **Rolle's Theorem:** If a function $f[a, a+h] \rightarrow R$ is such that:

- (1) It is continuous on $[a, a+h]$.
- (2) It is derivable on $(a, a+h)$.
- (3) $f(a) = f(a+h)$

Then there exist at least one $c \in (0,1)$ such that $f'(a+ch) = 0$.

- **Geometrical Interpretation of Rolle's Theorem:** Let $f[a,b] \rightarrow R$ be a function satisfying the three conditions of Rolle's theorem. So the graph $y = f(x)$ is such that:

- (1) It is continuous curve from the point $A(a, f(a))$ to the point $B(b, f(b))$.
- (2) It is having unique tangent at every intermediate point between A and B.
- (3) The ordinates $f(a), f(b)$ at the end points A and B of the curve are equal. Then there exists atleast one point $C(c, f(c))$ between A and B on the curve at which the tangent line is parallel to x-axis for some $c \in (a,b)$.

- **Algebraic Interpretation of Rolle's Theorem:** If $f(x)$ is a polynomial with a, b roots of $f(x) = 0$ i.e. $f(a) = f(b) = 0$, then $f(x)$ satisfies all the three conditions of Rolle's theorem. Therefore, there exist atleast one point $c \in (a,b)$ such that $f'(c) = 0$ i.e. c is a root of $f'(x) = 0$. So, if $f(x)$ is a polynomial, then between any two roots of $f(x) = 0$, there always lie a root of $f'(x) = 0$.

- **Lagrange's Mean Value Theorem:** If a function $f[a,b] \rightarrow R$ is such that:

- (1) It is continuous on $[a, b]$.
- (2) It is derivable on (a, b) .

Then there exist at least one $c \in (a,b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

OR

- **Lagrange's Mean Value Theorem:** If a function $f[a, a+h] \rightarrow R$ is such that:

- (1) It is continuous on $[a, a+h]$.
- (2) It is derivable on $(a, a+h)$.

Then there exist at least one $c \in (0,1)$ such that $f'(a+ch) = \frac{f(a+h) - f(a)}{h}$.

- **Geometrical Interpretation of Lagrange's Mean Value Theorem:** Let $f[a,b] \rightarrow R$ be a function satisfying the two conditions of Lagrange's theorem. So the graph $y = f(x)$ is such that:

- (1) It is continuous curve from the point A(a, f(a)) to the point B(b, f(b)).
- (2) It is having unique tangent at every intermediate point between A and B.
- (3) There exists atleast one point C(c, f(c)) between A and B on the curve at which the tangent line is parallel to chord AB for some $c \in (a, b)$.

• **INCREASING, DECREASING, MAXIMA & MINIMA:**

- Let $f : A \rightarrow R$ be a real function

- (1) If $x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2) \quad \forall x_1, x_2 \in A$ then f is said to be increasing in A.
- (2) If $x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2) \quad \forall x_1, x_2 \in A$ then f is said to be decreasing in A.
- (3) If f is either increasing or decreasing in A then f is said to be monotonic function in A.

- Let the real function f be defined in a neighborhood of 'a'.

- (1) If there exist $\phi > 0$, such that $\begin{cases} a - \phi < x < a \Rightarrow f(x) < f(a) \\ a < x < a + \phi \Rightarrow f(x) > f(a) \end{cases}$ then f is said to be increasing at 'a'.
- (2) If there exist $\phi > 0$, such that $\begin{cases} a - \phi < x < a \Rightarrow f(x) > f(a) \\ a < x < a + \phi \Rightarrow f(x) < f(a) \end{cases}$ then f is said to be decreasing at 'a'.

- Let the real function f be differentiable at 'a'.

- (1) If $f'(a) > 0$ then f is increasing at 'a'.
- (2) If $f'(a) < 0$ then f is decreasing at 'a'.
- (3) The converse of the above is not true.

If f is increasing or decreasing at 'a' then $f'(a)$ may be zero.

Example: $f(x) = x^3$ at $x = 0$.

- If $f(x)$ is increasing at a point 'a' then $-f(x)$ is decreasing at that point 'a'.

- Let f be a real function which is differentiable at 'a'.

- (1) If $f'(a) = 0$ then we say that $f(x)$ is stationary at $x = a$.
- (2) $(a, f(a))$ is called as stationary point.
- (3) $f(a)$ is called as stationary value.

- Let the real function f be defined in a neighborhood of 'a'.

- (1) If there exist $\phi > 0$, such that $f(x) \leq f(a)$ for every $a - \phi < x < a + \phi$ then f is said to have relative (local) maximum at 'a' $f(a)$ is called as relative (local) maximum value.

- (2) If there exist $\delta > 0$, such that $f(x) \geq f(a)$ for every $a - \delta < x < a + \delta$ then f is said to have relative (local) minimum at 'a' $f(a)$ is called as relative (local) minimum value.
- The points at which a function attains either maximum or minimum are called Extreme points or Turning points of the function. Maximum or minimum values of functions are called as Extreme values or Turning values of the function.
 - **Test of Monotonicity:** Let $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) .
 - (1) If $f(x)$ is increasing on $[a, b]$ then $f'(x) \geq 0$, for all $x \in (a, b)$.
 - (2) If $f(x)$ is decreasing on $[a, b]$ then $f'(x) \leq 0$, for all $x \in (a, b)$.
 - (3) If $f'(x) > 0$ for all $x \in (a, b)$ then $f(x)$ is strictly increasing.
 - (4) If $f'(x) < 0$ for all $x \in (a, b)$ then $f(x)$ is strictly decreasing.
 - (5) If a function $f(x)$ is defined on (a, b) and $f'(x) > 0$ for all $x \in (a, b)$ except for a finite number of points, where $f'(x) = 0$, then $f(x)$ is strictly increasing on (a, b) .
 - (6) If a function $f(x)$ is defined on (a, b) and $f'(x) < 0$ for all $x \in (a, b)$ except for a finite number of points, where $f'(x) = 0$, then $f(x)$ is strictly decreasing on (a, b) .
 - **Necessary condition for extreme values of a function:**
 - (1) If a function $f(x)$ has extreme values $f(a)$, then $f'(a) = 0$, if it exists. Geometrically it means that the tangent line at $x = a$ to the graph of $y = f(x)$ is parallel to x-axis.
 - (2) $f(x) = |x|$ attains a minimum value at the origin even though its derivative at $x = 0$ does not exist.
 - **Sufficient condition for extreme values of a function:** Let $f(x)$ be derivable at $x = a$.
 - (1) $x = a$ is a point of relative maximum of $f(x)$ if $f'(a)$ changes sign from +ve to -ve as x passes through the point $x = a$ from left neighborhood to right neighborhood.
 - (2) $x = a$ is a point of relative minimum of $f(x)$ if $f'(a)$ changes sign from -ve to +ve as x passes through the point $x = a$ from left neighborhood to right neighborhood.
 - (3) If $f'(a)$ has the same sign in the entire neighborhood of $x = a$, then $x = a$ is not extremum.
 - **Sufficient condition for extreme values of a function:** Let $f(x)$ be derivable at $x = a$ and $f''(a)$ exists and is non zero.
 - (1) $f'(a) = 0$ and $f''(a) < 0 \Rightarrow x = a$ is a point of relative maximum.
 - (2) $f'(a) = 0$ and $f''(a) > 0 \Rightarrow x = a$ is a point of relative minimum.

• **Working method for getting extreme values of a function:**

Let $y = f(x)$ be a given function.

Step 1: Find $\frac{dy}{dx} = f'(x)$ and $\frac{d^2y}{dx^2} = f''(x)$

Step 2: Find all values of x for which $\frac{dy}{dx} = f'(x) = 0$

Step 3: If for any value of x from (2) $\frac{d^2y}{dx^2} = f''(x) < 0$, then that value of x is a point of maximum.

Step 4: If for any value of x from (2) $\frac{d^2y}{dx^2} = f''(x) > 0$, then that value of x is a point of minimum.

- If $f(x)$ is increasing in $[a, b]$ then $f(a)$ is the minimum value and $f(b)$ is the maximum value of $f(x)$ in $[a, b]$.
- If $f(x)$ is decreasing in $[a, b]$ then $f(a)$ is the maximum value and $f(b)$ is the minimum value of $f(x)$ in $[a, b]$.
- If $f(x) = x^2 + k$, where k is the real number, then $f(x) \geq k, \forall x \in R$. Hence minimum value of $f(x) = k$.
- If $f(x) = -x^2 + k$, where k is the real number, then $f(x) \leq k, \forall x \in R$. Hence maximum value of $f(x) = k$.

• **Some Important points:**

(1) The minimum value of $f(x) = a \cos x + b \sin x + c$ is $-\sqrt{a^2 + b^2} + c$.

(2) The maximum value of $f(x) = a \cos x + b \sin x + c$ is $\sqrt{a^2 + b^2} + c$.

(3) The minimum value of $f(x) = a \cos^2 x + b \sin^2 x + c$ is $b + c$, here $a > b$.

(4) The maximum value of $f(x) = a \cos^2 x + b \sin^2 x + c$ is $a + c$, here $a > b$.

(5) The minimum value of $f(x) = a \tan x + b \cot x + c$ is $2\sqrt{ab} + c$ attains at $\tan x = \sqrt{\frac{b}{a}}$.

(6) The minimum value of $f(x) = a \sec x + b \operatorname{cosec} x + c$ is $(a^{\frac{2}{3}} + b^{\frac{2}{3}})^{\frac{3}{2}} + c$ attains at $\tan x = \left(\frac{b}{a}\right)^{\frac{1}{3}}$.

(7) The minimum value of $f(x) = \sin^m x \cdot \cos^n x$ is $\sqrt{\frac{m^m n^n}{(m+n)^{m+n}}}$ attains at $\tan x = \sqrt{\frac{m}{n}}$.

(8) The sum of two numbers is k . If the sum of their squares is minimum, then the numbers are $\frac{k}{2}, \frac{k}{2}$.

- (9) The sum of two numbers is k and the least sum of their squares is $\frac{k^2}{2}$.
- (10) The sum of two numbers is k . If their product is maximum, then the numbers are $\frac{k}{2}, \frac{k}{2}$.
- (11) The product of two numbers is k . If the sum of their squares is minimum, then the numbers are $\sqrt{k}, \sqrt{k}$.
- (12) The sum of two numbers is k . If the product of the squares of the first and the cube of the second is maximum then the numbers are $\frac{2k}{5}, \frac{3k}{5}$.
- (13) If $a > 0, b > 0, x > 0$, the least value of $f(x) = ax + \frac{b}{x}$ is $2\sqrt{ab}$.
- (14) If $x > 0$, the least value of $f(x) = x + \frac{1}{x}$ is 2 .
- (15) The maximum rectangle inscribed in a circle is a square.
- (16) The maximum area of the rectangle in a circle of radius r is $2r^2$.
- (17) The maximum triangle inscribed in a circle is an equilateral triangle.
- (18) The maximum area of the triangle in a circle of radius r is $\frac{3\sqrt{3}}{4}r^2$ sq. units.
- (19) The perimeter of a sector C cm. The maximum area of the sector is $\frac{C^2}{16}$ sq. cm.
- (20) The perimeter of a sector is given. The area of sector is maximum. Then angle of sector is 2 radian.
- (21) The area of sector is ' k ' sq cm. Then the least perimeter of sector is $4\sqrt{k}$ cm.
- (22) The hypotenuse of a right angled triangle is ' k '. If the area of the triangle is maximum, then the sides are $\frac{k}{\sqrt{2}}, \frac{k}{\sqrt{2}}$.
- (23) Two sides of a triangle are given. The area of triangle is maximum, then the angle between the sides is $\frac{\pi}{2}$.
- (24) The sum of hypotenuse and one side of right angled triangle is given. The area is maximum, then angle between them is $\frac{\pi}{3}$.
- (25) The area of greatest rectangle inscribed in an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $2ab$.
- (26) An open box of maximum volume is made from a square piece of tin of side ' k ' by cutting for equal squares pieces from four corners and folding up the tin, then length of square cut is $\frac{k}{6}$.

- (27) An open box of maximum volume is made from a rectangular piece of tin of length 'a' and breadth 'b' by cutting for equal squares pieces from four corners and folding up the tin, then length of square cut is $\frac{(a+b) - \sqrt{a^2 + b^2 - ab}}{6}$.

• **Concavity & Convexity:**

- (1) Let P be a point on the curve $y = f(x)$, where the tangent PT is not parallel to y-axis. The curve is said to be concave upwards (or convex downwards) at P if in the immediate neighborhood of P, the curve lies above the tangent PT on both sides.
- (2) Let P be a point on the curve $y = f(x)$, where the tangent PT is not parallel to y-axis. The curve is said to be concave downwards (or convex upwards) at P if in the immediate neighborhood of P, the curve lies below the tangent PT on both sides.

• **Criteria for Concavity & Convexity:** At a point P on the curve $y = f(x)$, the curve is:

- (1) Convex Downward if $f''(x) > 0$.
- (2) Concave Downward if $f''(x) < 0$.

• **Point of Inflexion:** A point on the curve at which the curve changes from concavity to convexity or vice-versa is called as point of inflexion. At a point of inflexion, the tangent to the curve crosses the curve.

• **Criteria for point of Inflexion:** A point P on the curve is a point of inflexion if:

- (1) $f''(x) = 0$ at P.
- (2) $f''(x)$ Changes its sign through this point P.
